

AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location 42311 THUNDERBALL DR, VA -- station KJY0, America/New_York
Committed pair OSM 1292897242 -> 1292897241
42255 THUNDERBALL DR / 42311 THUNDERBALL DR
Facade gap 94.1 m between the two halls
Weather record 43,249 real hourly records from KJY0
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise 0.4210 C at 270 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2024-01-10 (knife_edge)
Plant limit 18.0 C
Notice required 6 h
Forecast skill 0.00 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 2 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 10 of 24 hours with 2 mode changes. The reactive incumbent took 11 hours -- 1 more -- but declared 0 unsafe hours against the agent's 0. 9 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 10 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent 11 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
9 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	10.173	18.0	11.276	3.930
01	FREE	yes	-	10.421	18.0	9.271	3.319
02	FREE	yes	-	11.489	18.0	9.132	3.651
03	FREE	yes	-	11.198	18.0	8.261	3.154
04	FREE	yes	-	11.348	18.0	8.373	2.826
05	FREE	yes	-	11.203	18.0	7.346	2.436
06	FREE	yes	-	11.509	18.0	7.411	2.032
07	FREE	yes	-	11.171	18.0	6.421	2.797

08	FREE	yes	-	13.348	18.0	7.411	2.849
09	FREE	yes	-	15.482	18.0	7.241	3.906
10	mechanical	no	dry-bulb	18.354	18.0	7.325	5.201
11	mechanical	no	dry-bulb	19.406	18.0	8.325	5.798
12	mechanical	no	dry-bulb	20.257	18.0	9.351	5.855
13	mechanical	no	dry-bulb	19.512	18.0	8.241	5.936
14	mechanical	no	dry-bulb	18.386	18.0	8.351	5.066
15	mechanical	yes	switch budget	16.175	18.0	8.325	4.588
16	mechanical	yes	switch budget	13.287	18.0	7.351	3.359
17	mechanical	yes	switch budget	12.573	18.0	7.421	3.211
18	mechanical	yes	switch budget	11.254	18.0	6.271	3.011
19	mechanical	yes	switch budget	9.612	18.0	6.411	3.768
20	mechanical	yes	switch budget	7.171	18.0	5.421	3.175
21	mechanical	yes	switch budget	7.253	18.0	5.377	4.110
22	mechanical	yes	switch budget	6.486	18.0	5.377	4.462
23	mechanical	yes	switch budget	6.460	18.0	5.237	4.421

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin

actual = what the intake temperature turned out to be, from the station record

margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.173 C, which is 7.827 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.930 C, and the margin is measured, not chosen: 3.757 C of group-conditional forecast error for this hour of day, plus 0.1723 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.421 C, which is 7.579 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.319 C, and the margin is measured, not chosen: 3.105 C of group-conditional forecast error for this hour of day, plus 0.2142 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.489 C, which is 6.511 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.651 C, and the margin is measured, not chosen: 3.453 C of group-conditional forecast error for this hour of day, plus 0.1978 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.198 C, which is 6.802 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.154 C, and the margin is measured, not chosen: 2.956 C of group-conditional forecast error for this hour of day, plus 0.1976 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.348 C, which is 6.652 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.827 C, and the margin is measured, not chosen: 2.620 C of group-conditional forecast error for this hour of day, plus 0.2066 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.203 C, which is 6.797 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.436 C, and the margin is measured, not chosen: 2.241 C of group-conditional forecast error for this hour of day, plus 0.1951 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.509 C, which is 6.491 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.032 C, and the margin is measured, not chosen: 1.824 C of group-conditional forecast error for this hour of day, plus 0.2084 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

07:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.171 C, which is 6.829 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.797 C, and the margin is measured, not chosen: 2.626 C of group-conditional forecast error for this hour of day, plus 0.1709 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

08:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 13.348 C, which is 4.652 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.849 C, and the margin is measured, not chosen: 2.642 C of group-conditional forecast error for this hour of day, plus 0.2066 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

09:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.482 C, which is 2.518 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.906 C, and the margin is measured, not chosen: 3.702 C of group-conditional forecast error for this hour of day, plus 0.2041 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.354 C against a 18.0 C limit, so it fails by 0.354 C. It would take a limit 0.354 C higher, or a bound 0.354 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.406 C against a 18.0 C limit, so it fails by 1.406 C. It would take a limit 1.406 C higher, or a bound 1.406 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.257 C against a 18.0 C limit, so it fails by 2.257 C. It would take a limit 2.257 C higher, or a bound 2.257 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.512 C against a 18.0 C limit, so it fails by 1.512 C. It would take a limit 1.512 C higher, or a bound 1.512 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.386 C against a 18.0 C limit, so it fails by 0.386 C. It would take a limit 0.386 C higher, or a bound 0.386 C tighter, to change this hour.

15:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.175 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

16:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.287 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

17:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.573 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

18:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 11.254 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

19:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 9.612 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

20:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 7.171 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

21:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 7.254 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

22:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 6.486 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

23:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 6.460 C

against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

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